THE NEW WEAPON AGAINST TB

With India running against time to eliminate tuberculosis by 2025, a new drug regimen offers hope. But persistent shortages of even basic medications threaten to derail progress and fuel drug resistance

By **SONALI ACHARJEE**



VITAL SCAN
A doctor checking a TB patient's chest X-ray at the Médecins Sans Frontières clinic in Mumbai

AFP

For 67-year-old Faizan Khan, who used to work as a plumber in Mumbai suburb Thane, swallowing multiple pills for the treatment of tuberculosis was a living nightmare. But a friend of his had developed drug-resistant or DR-TB and succumbed to it after he could not cope with the extensive medication. And so Khan persevered with his own treatment.

Hope flared for Khan and others suffering like him in mid-August, as the Government of India approved the rollout of a new TB regimen. Of a shorter duration—six months—and comprising just three oral drugs, the BPaL line of treatment hopes to potentially improve patient compliance and hence reduce the risk associated with MDR (multi-drug resistant) TB. However, twice in less than a year—August 2023 and March 2024—there have been mass-scale reports of drug shortages for weeks. And halting treatment due to the unavailability of drugs increases the risk of TB becoming resistant to these drugs. So, what is the way out?

A Killer Bacterium

A bacterial infection caused by the tenacious Mycobacterium tuberculosis, TB primarily targets the lungs, turning even a slight cough, sneeze or spit into a potential carrier of the disease. It typically causes persistent cough, fever, night sweats and weight loss. If left untreated, TB can lead to serious complications such as lung damage, meningitis or infection of other organs. India has the highest TB burden in the world, accounting for nearly 27 per cent of all global cases in 2022, as per the World Health Organization (WHO) Global TB report 2023. The same year, India lost 331,000 lives to the disease, a fourth of the 1.3 million deaths reported globally. In 2023, 2.55 million cases were notified in the country, according to the India TB Report 2024. While the bulk of these cases were of DS (drug-sensitive) TB, which responds to all first-line TB medications, 63,939 were of MDR-TB—caused by a strain resistant to at least two of the most potent first-line drugs (Isoniazid and Rifampicin). Then there is extensively drug-resistant TB—a rare type of MDR-TB caused by a strain resistant to even most of the second-line medications.

"Bacteria are extremely smart and have been around for millions of years," says Anirban Mahapatra, microbiologist and author of *When the Drugs Don't Work*. "Misuse [of medication] or incomplete courses of antibiotics have resulted in them learning how to outsmart several key drugs in TB treatment. If we don't take steps to tackle drug resistance, we could very well be in a situation where we run out of treatment options for TB."

Currently, newly diagnosed drug-sensitive (DS) TB patients are given a four-drug fixed dose combination (4FDC) for two months followed by a 3FDC for another two months. The regimen for MDR-TB comprises 11 drugs given over nine months (see *TB Treatment and Challenges*). However, according to Dr Bornali Dutta, director of respiratory medicine at Medanta, Gurugram, treatment typically lasts for a year on average and can also take up to two years in a few cases.